Oliver Braunsdorf

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SUMMARY

Self-driven IT security researcher and software engineer with a team-first attitude. Main field of interest is software security, including static and dynamic program analysis, program hardening, and secure software design—especially with the Rust programming language. Passionate learner, communicative problem solver, open-source software creator, and American Football player and coach.

PROFESSIONAL EXPERIENCE

Fraunhofer Institute for Applied and Integrated Security (AISEC) Munich, Germany
Research Associate, Department Secure Operating Systems Jan 2018 – Present

- Consulted Fortune 500 companies and federal agencies in automotive, aviation, and critical infrastructures, including threat modeling, software architecture, and code review.
- Managed research and software engineering projects, leading teams of 2-4 people and presented results to stakeholders with different levels of technical depth.
- Created static and dynamic analysis tools based on LLVM and the Rust compiler for detecting and preventing software vulnerabilities.

Secunet Security Networks AG Dresden, Germany
Internship, Research & Development Apr 2016 – Sept 2016

- Implemented optimization strategies inside the Linux Kernel's IPsec subsystem to maximize throughput of encrypted connections in gigabit ethernet networks.

Dotsource GmbH Jena, Germany
Internship, Software Engineer Quality Assurance Nov 2010 – June 2011

- Increased testing efficiency by developing an infrastructure for automated browser-based UI tests of web shops which reduced the need for manual testing.

EDUCATION

Technical University of Munich Munich, Germany
Ph.D. Candidate in Computer Science Jan 2019 – Present

- Research directions: compiler-based software security, memory safety, control-flow integrity, static analysis, programming languages

Technical University of Ilmenau Ilmenau, Germany
M.Sc. in Computer Science, final grade: 1.2 (GPA: 3.8) Apr 2015 – Oct 2017

- Honors: Degree with distinction
- Specializations: IT-Security, Programming Languages
- Thesis Topic: *Static Verification of Secure Design Patterns*

Technical University of Ilmenau Ilmenau, Germany
B.Sc. in Computer Science, final grade: 1.7 (GPA: 3.3) Oct 2011 – Mar 2015

- Specializations: IT-Security, Database Systems, Distributed Systems
- Thesis Topic: *Incremental Clustering Algorithms for Analyzing Monitoring Data Streams of Overlay Networks*

LANGUAGES, TECHNICAL SKILLS & CERTIFICATIONS

Languages:	German (native), English (fluent), Spanish (beginner)
Programming Languages:	Rust, C/C++, Python (used daily for work and research)
Compiler Development:	LLVM Framework, Rust Compiler
Debugging & Profiling:	GDB, Valgrind, Linux perf, Hyperfine
Build Tools & Processes:	GitHub Actions, Gitlab CI, Docker, CMake, Cargo
Certifications:	Professional Scrum Master I, Scrum.org

OPEN-SOURCE SOFTWARE PROJECTS

Industrial Dataspace Communication Protocol (Rust)
Designed and implemented a P2P network protocol to establish secure communication channels based on mutual remote attestation. Used the TLA+ model checker to model the protocol's state machine and proof security and liveness properties. Refactored implementation to utilize async Rust for performance.
https://github.com/International-Data-Spaces-Association/idscp2-rust

Codyze (Java)
Collaborated in a team of scientists to create a domain-specific language to formally express cryptographic guidelines and a static analysis framework to verify correct crypto API usage in applications.
https://github.com/Fraunhofer-AISEC/codyze

Playbook Creator (C++/Rust)
Maintaining an open-source project which provides a QT-based GUI application for creating American Football playbooks. Communicating with football teams across Europe to improve its features.
https://github.com/obraunsdorf/playbook-creator

RESEARCH PAPERS

Braunsdorf, O., Sessinghaus, S., & Horsch, J. (2022). Resilient CFI: Compiler-based Attack Origin Tracking with Dynamic Taint Analysis. In *International Conference on Information Security and Cryptology* (pp. 175-191). Springer, Cham.

Pöhn, D., Wessel, S., Fischer, F., Braunsdorf, O., Wenninger, F., Seydel, D., & Roscher, K. (2018). A Rapid Innovation Framework for Connected Mobility Applications. [White paper]

INTERESTS

Defensive captain on the German National Team for American Flag Football since 2018.

Player in various team and individual sports, e.g., American Football, Soccer, Volleyball, and Pickleball.

Programmer of GUI application software for sports teams.